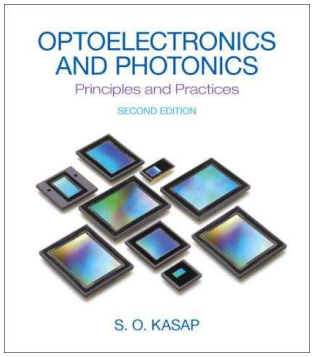
Chapter 3 Semiconductor Science and Light-Emitting Diodes



- Ref-2: Ch.1: Basic Semiconductor Properties

3.1 Review of Semiconductor Concepts and Energy Bands

- Ref-2: Ch.3-A: Energy Band Theory
- Ref-2: Ch.3-B: Energy Band Theory

3.4 Direct and Indirect Bandgap Semiconductors: E-k Diagrams

3.2 Semiconductor Statistics

3.3 Extrinsic Semiconductors

- Ref-2: Ch.4 : Equilibrium Carrier Statistics

3.5 pn Junction Principles---중간고사---

3.6 pn Junction Reverse Current

3.7 pn Junction Dynamic Resistance and Capacitances

3.9 pn Junction Band Diagram

3.8 Recombination Lifetime

Ref-2:Ch5: Recombination-Generation Processes-1/2

Shockley–Read–Hall (SRH) recombination

3.10 Heterojunctions-1,**2**

3.11 Light-Emitting Diodes: Principles

3.12 Quantum Well High Intensity LEDs

3.13 LED Materials and Structures

3.14 LED Efficiencies and Luminous Flux

3.15 Basic LED Characteristics

3.16 LEDs for Optical Fiber Communications

3.17 Phosphors and White LEDs

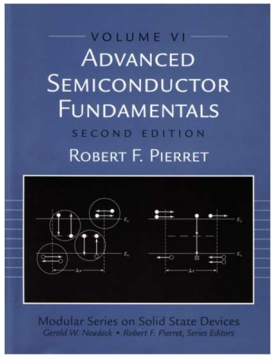
+ Advanced Semiconductor Fundamentals:
2nd Ed. Rober F. Peirret

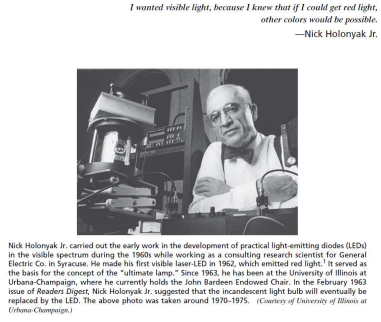
Ch.1: Basic Semiconductor Properties

Ch.3: Energy Band Theory

Ch.4: Equilibrium Carrier Statistics

Ch 5: Recombination-Generation Processes





I wanted visible light, because I knew that if I could get red light, other colors would be possible.
—Nick Holonyak Jr.

Nick Holonyak Jr. carried out the early work in the development of practical light-emitting diodes (LEDs) in the visible spectrum during the 1960s while working as a consulting research scientist for General Electric Co. in Syracuse. He made his first visible laser-LED in 1962, which emitted red light.[1] It served as the basis for the concept of the "ultimate lamp." Since 1963, he has been at the University of Illinois at Urbana-Champaign, where he currently holds the John Bardeen Endowed Chair. In the February 1963 issue of *Readers Digest*, Nick Holonyak Jr. suggested that the incandescent light bulb will eventually be replaced by the LED. The above photo was taken around 1970–1975. *(Courtesy of University of Illinois at Urbana-Champaign.)*

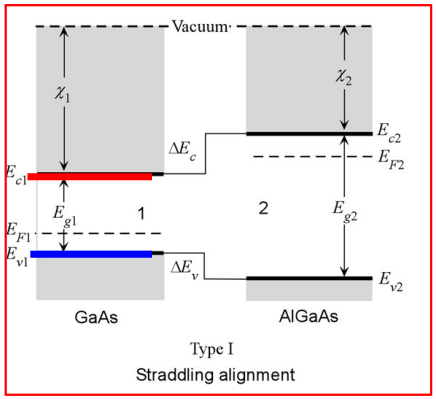
3.10 Heterojunctions: *Np* Heterojunction Energy Band Diagram

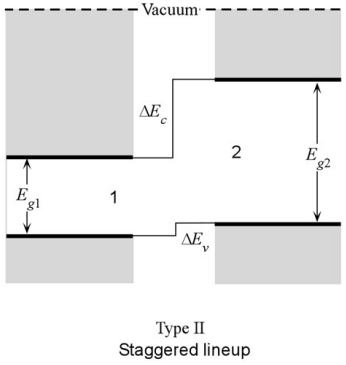
➢ Type I: Straddled bandgap alignment heterojunction

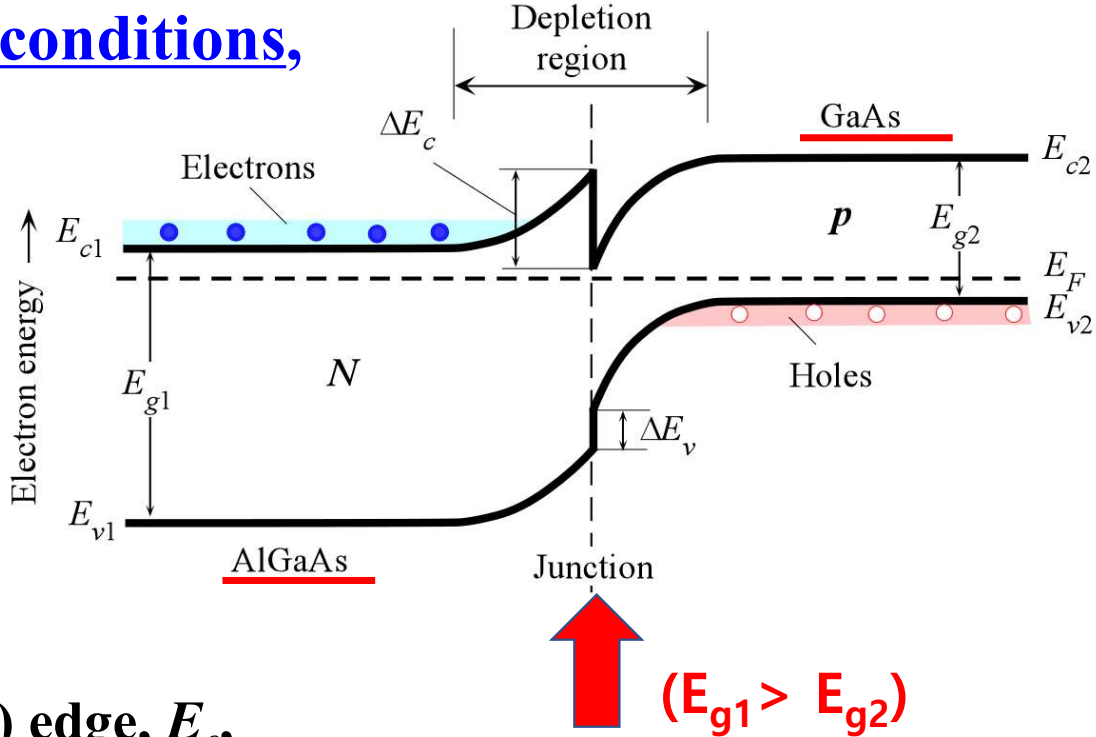
- Under open circuit & equilibrium conditions,

→ Fermi level E_F must be uniform

→ Continuous throughout the device







- **If E_F is close to the conduction band (CB) edge, E_c,**

  → Results in an ***n*-type**

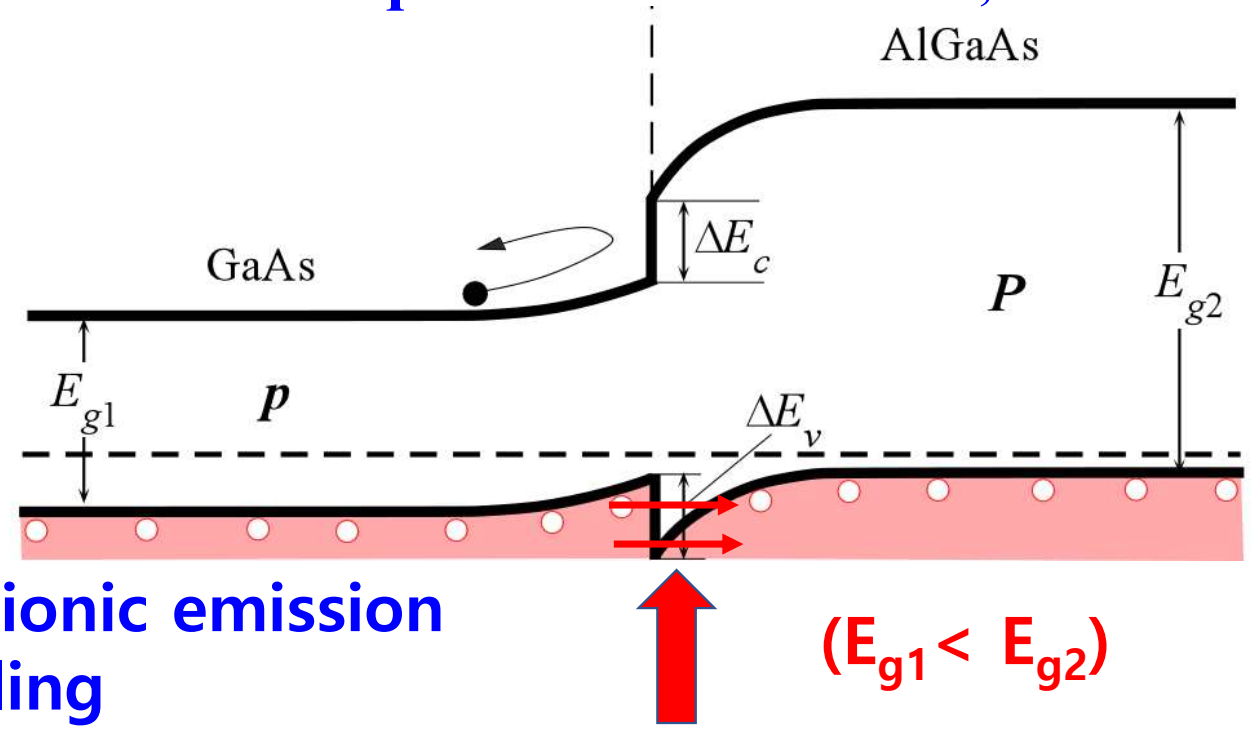
- **If it is close to the valence band (VB) edge, E_v,**

  → Results in a ***p*-type semiconductor.**

- **Discontinuity ΔE_c in E_c, & ΔE_v in E_v, right at the junction**

pP Heterojunction Energy Band Diagram

- **Under open circuit & equilibrium conditions,**



+ Thermionic emission
+ Tunneling

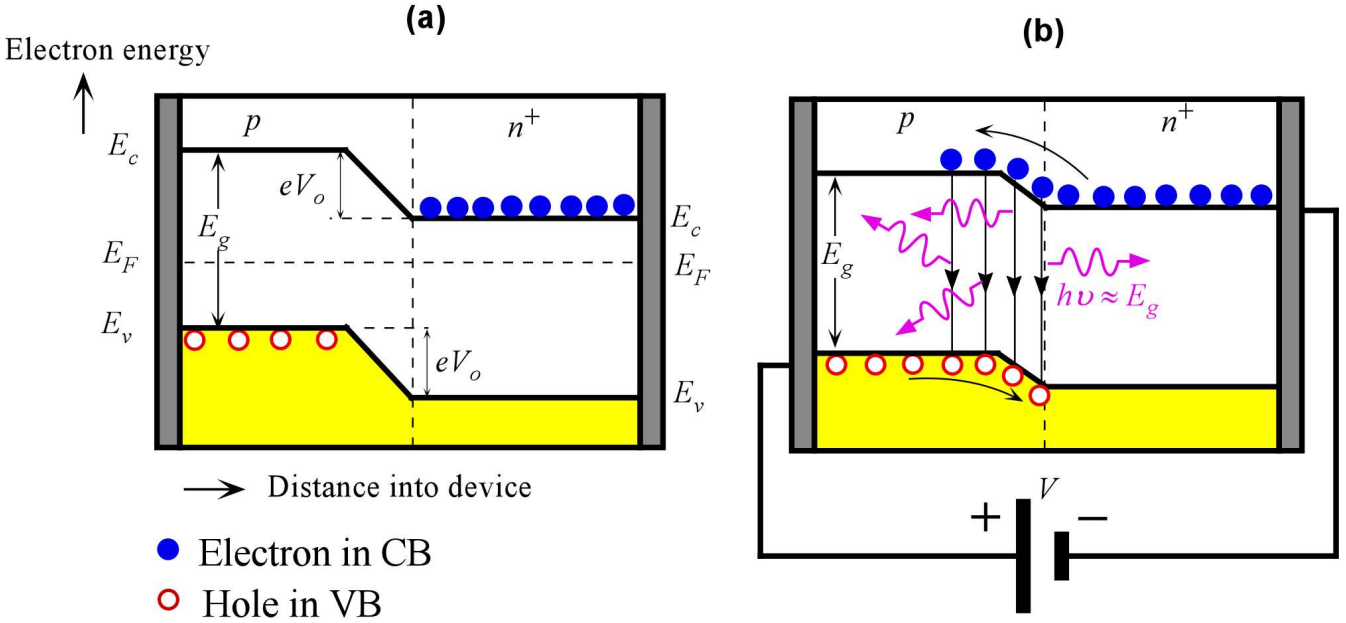
($E_{g1} < E_{g2}$)

Discontinuity
ΔE_c in E_c,
& ΔE_v in E_v,
right at the junction

3.11 Light-Emitting Diodes (LEDs): Principles

A. Homojunction LEDs

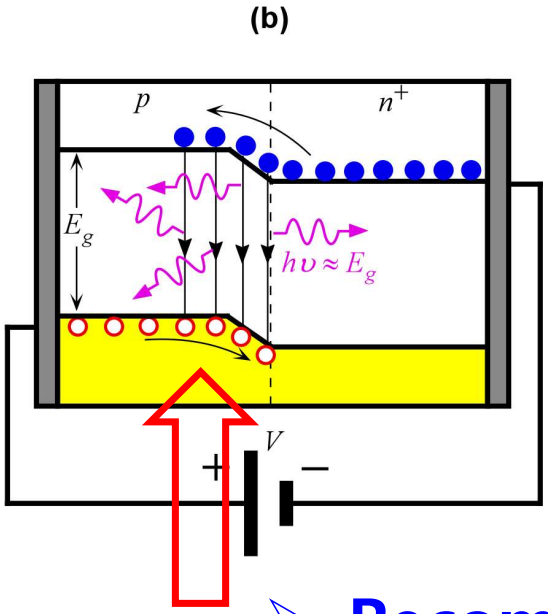
- **LED – a *pn* junction**
- Typically (Not always) made from a **direct** bandgap semiconductor
- An e-h pair recombination ➔ Emission of a photon
- **Emitted photon energy**
  ~ Bandgap energy : hυ ~ E_g



(a) The energy band diagram of a *pn*⁺ (heavily *n*-type doped) junction **without any bias**

- **Built-in potential V_o** prevents electrons from diffusing from n^+ to *p* side.

Basic LED Principle



(b) **The applied bias (V) reduces V_o**

→Allows electrons to diffuse,
be injected, into the *p*-side.

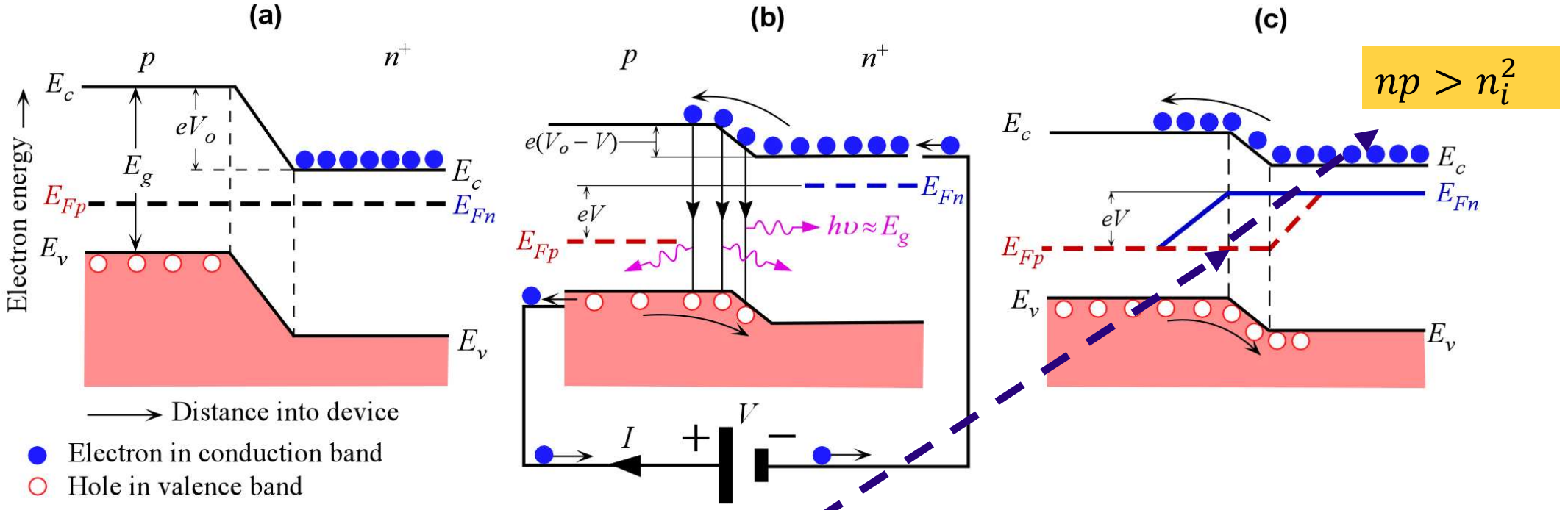
→**Recombination**
around the junction

+ within the diffusion length of the electrons
in the *p*-side **leads to photon emission**

- **Recombination zone: Active region**

- **Injection electroluminescence:**
  - **Spontaneous emission**: Emitted photons in random directions
    - Statistical nature of the natural recombination process
  - **Recombination rate** : **Depends on *np***
    → **Need to confine** the carriers
    → **increases the carrier concentration**
    → **Heterostructures**

Light Emitting Diodes



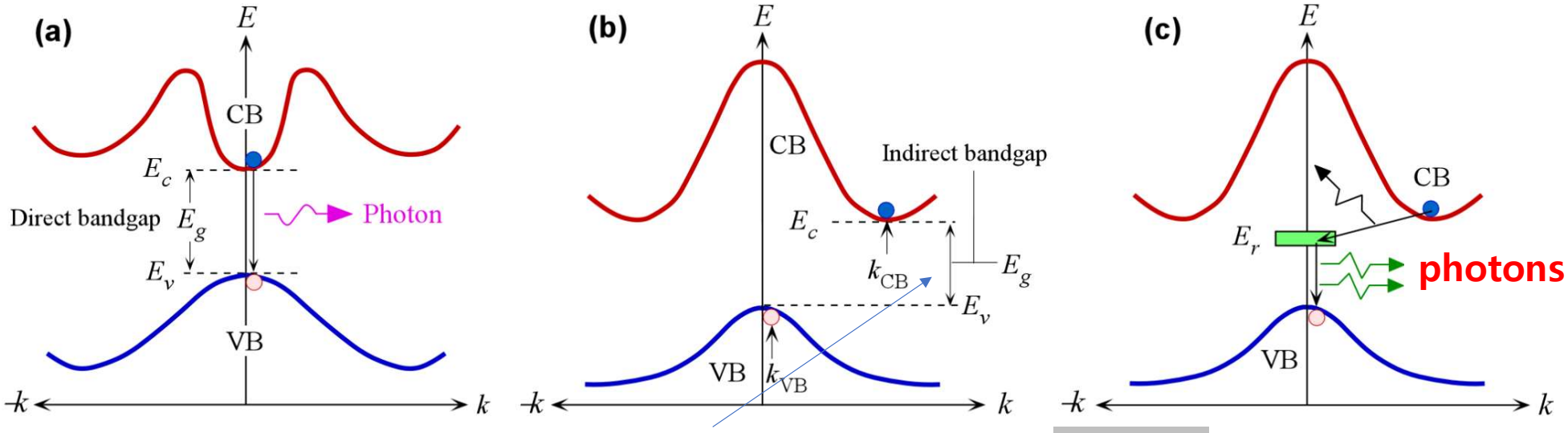
$np > n_i^2$

$$n = N_c \exp\left[-\frac{(E_c - E_{Fn})}{k_B T}\right]$$

$$p = N_v \exp\left[-\frac{(E_{Fp} - E_v)}{k_B T}\right]$$

(c) Quasi-Fermi levels E_{Fp} and E_{Fn} for holes and electrons across a forward biased *pn*-junction.

Light Emitting Diodes



GaP (2.26 eV @ RT): ID

~ 550 nm

GaP:N a recombination center

- **GaP:N**
  - -**Green** LED (**570 nm ~ 600 nm**)

N:P – Isoelectronic dopant

3.11 Light-Emitting Diodes: Principles

B. Heterostructure High Intensity LEDs

◆ **Problems with homojunction LEDs:**

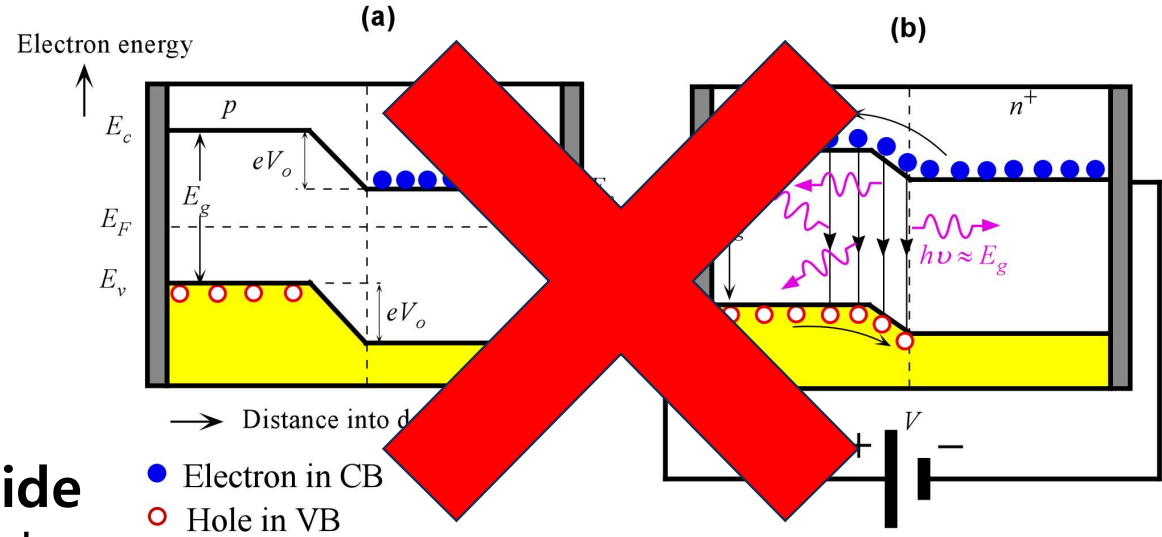
➔ **Low efficiency**

(1) p region must **be narrow**
to allow the photons **to escape**
without much reabsorption

(2) When p-side is narrow,
Some of injected electrons **in the p side**
reach the surface by diffusion and
recombine through crystal defects
near the surface

➔ **Radiationless recombination**

(3) **Reabsorption by the bulk materials**

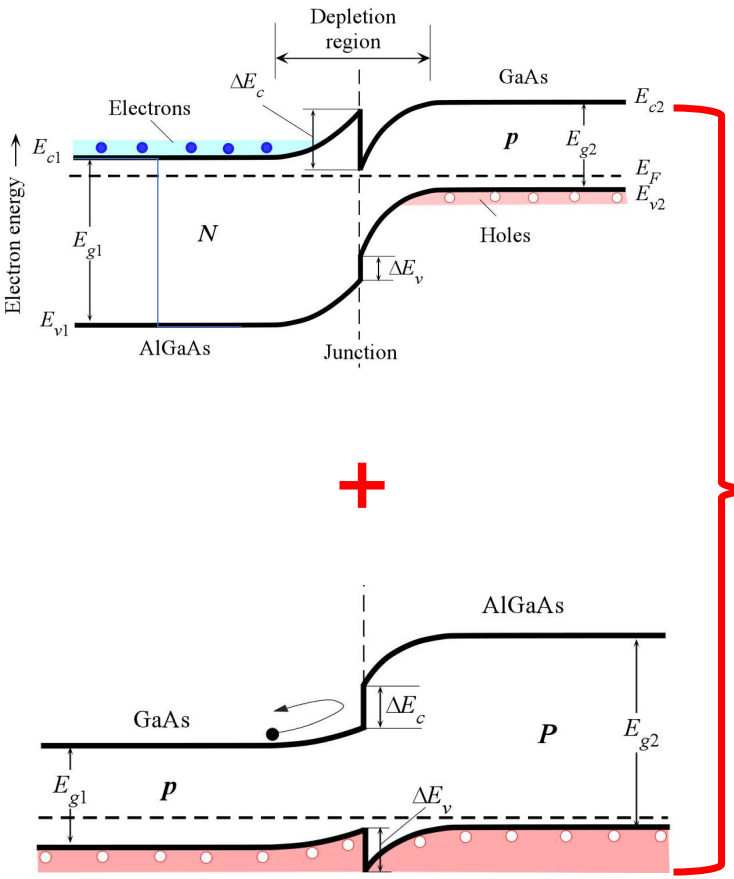


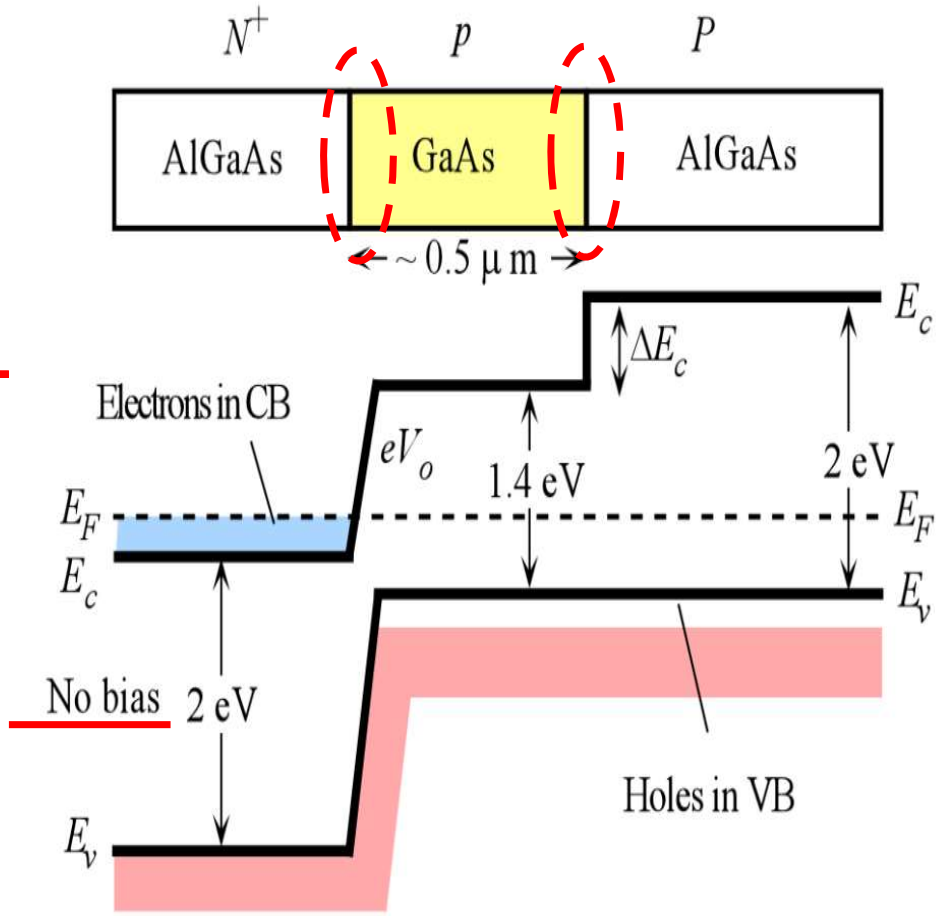
Then What ?

- **Heterostructure LEDs: High efficiency,**
- **Wider bandgap**

→ **Lower refractive index → Wave guide**

Double-Heterostructure

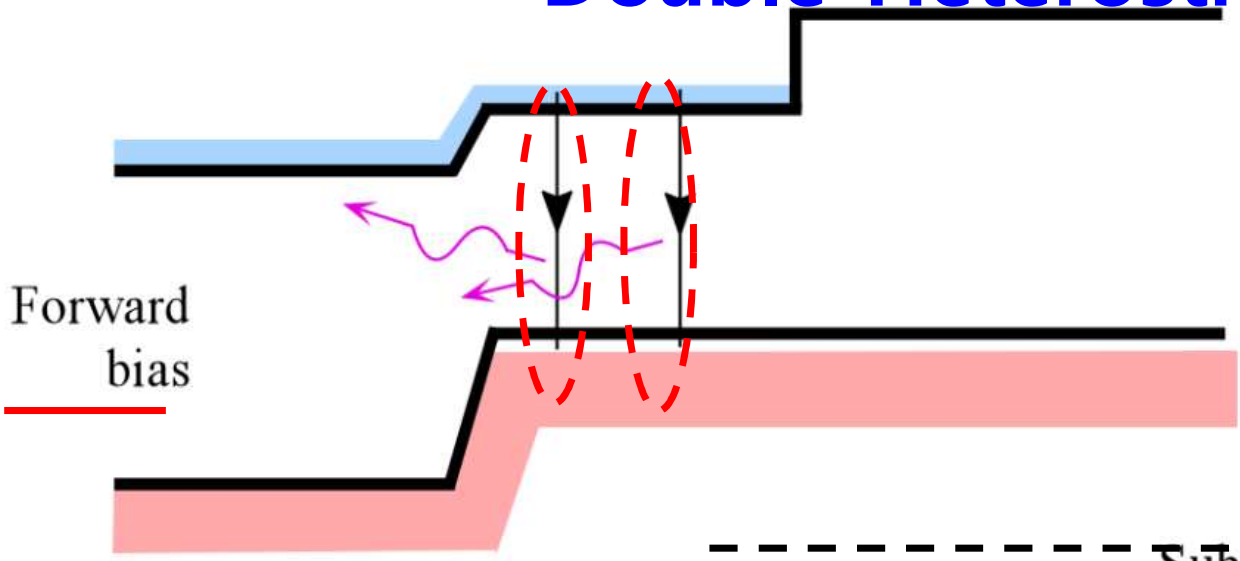




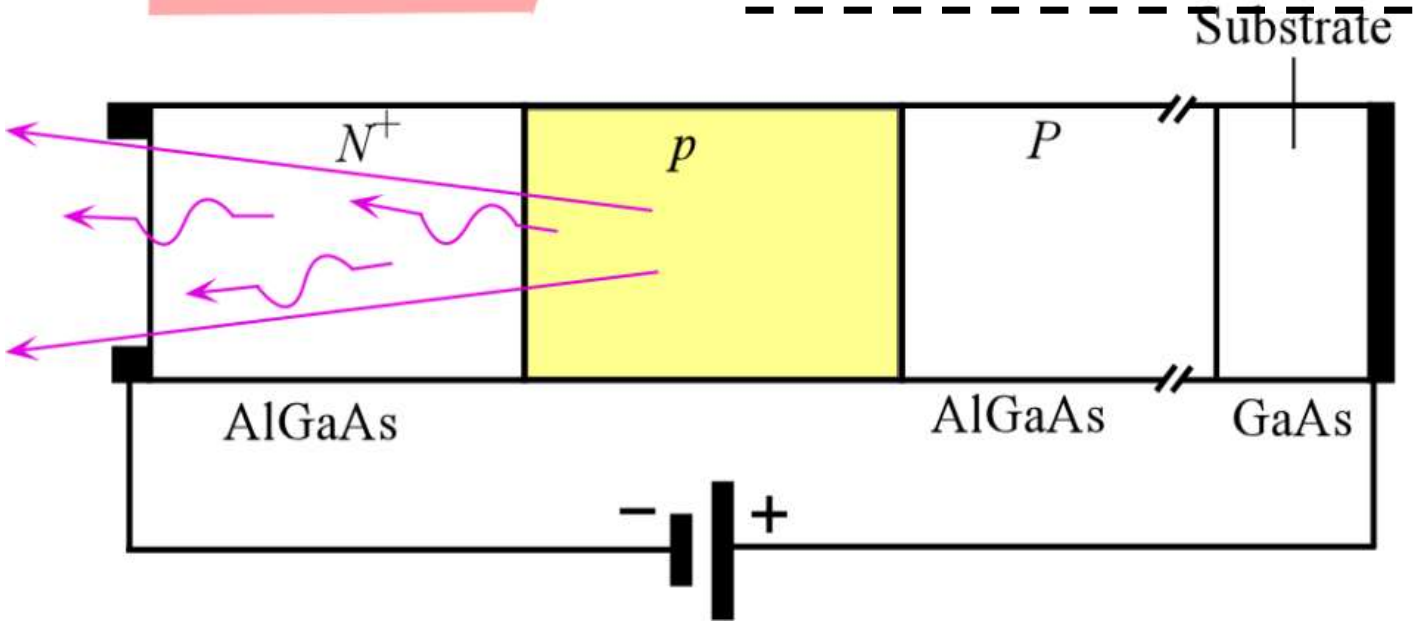
(a) A double heterostructure diode has two heterojunctions which are between two different bandgap semiconductors (GaAs and AlGaAs)

(b) A simplified energy band diagram with exaggerated features. E_F must be uniform

Double-Heterostructure

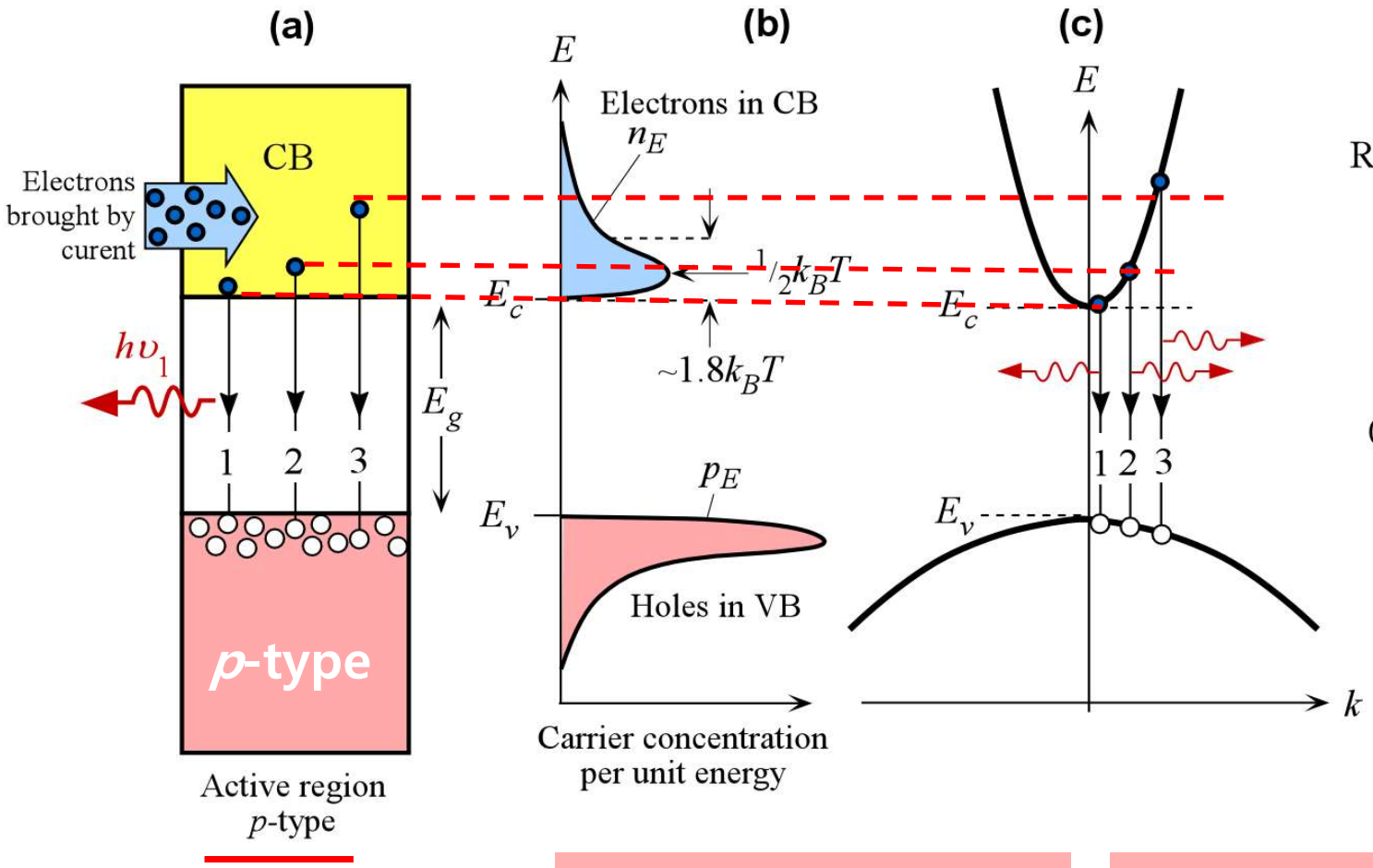


(c) Forward biased simplified energy band diagram



(d) Forward biased LED. Schematic illustration of photons escaping reabsorption in the AlGaAs layer and being emitted from the device. Layers are grown on a GaAs crystal substrate

C. Output Spectrum-Emission Spectrum

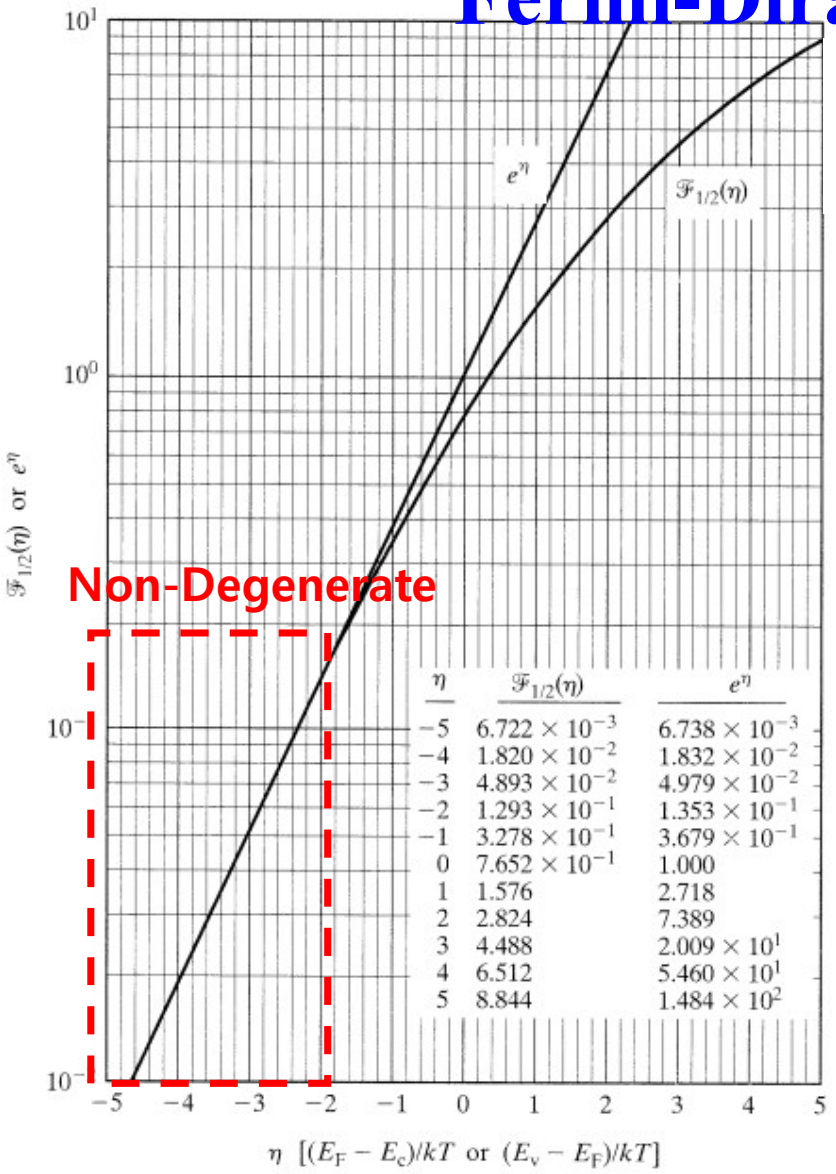


$$h\upsilon_o \approx E_g + \tfrac{1}{2}k_BT$$

$$h\upsilon_o \approx E_g + \tfrac{1}{2}k_BT \qquad h\Delta\upsilon = mk_BT$$

$$h\Delta\upsilon = mk_BT$$

Fermi-Dirac Distribution/Integral



η	$\mathcal{F}_{1/2}(\eta)$	e^η
−5	6.722×10^{-3}	6.738×10^{-3}
−4	1.820×10^{-2}	1.832×10^{-2}
−3	4.893×10^{-2}	4.979×10^{-2}
−2	1.293×10^{-1}	1.353×10^{-1}
−1	3.278×10^{-1}	3.679×10^{-1}
0	7.652×10^{-1}	1.000
1	1.576	2.718
2	2.824	7.389
3	4.488	2.009×10^1
4	6.512	5.460×10^1
5	8.844	1.484×10^2

$$n = \int_{E}^{E_{top}} g_c(E) f(E) dE$$

$$p = \int_{E_{bottom}}^{E_v} g_v(E)[1 - f(E)] dE$$

$$n = N_C \mathcal{F}_{1/2}(\eta_c)$$

$$p = N_V \mathcal{F}_{1/2}(\eta_v)$$

$$g(E) = 4\pi (2m_e)^{3/2} h^{-3} E^{1/2} = \mathbf{A}\, E^{1/2}$$

$$n = N_C e^{(E_F - E_c)/kT} \quad \ldots E_c - E_F \geq 3kT \ (\eta_c \leq -3)$$

Non-Degenerate

$$p = N_V e^{(E_v - E_F)/kT} \quad \ldots E_F - E_v \geq 3kT \ (\eta_v \leq -3)$$

$$f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

$$\eta_c = (E_F - E_c)/kT$$

$$\mathcal{F}_{1/2}(\eta) = \frac{2}{\sqrt{\pi}} F_{1/2}(\eta)$$

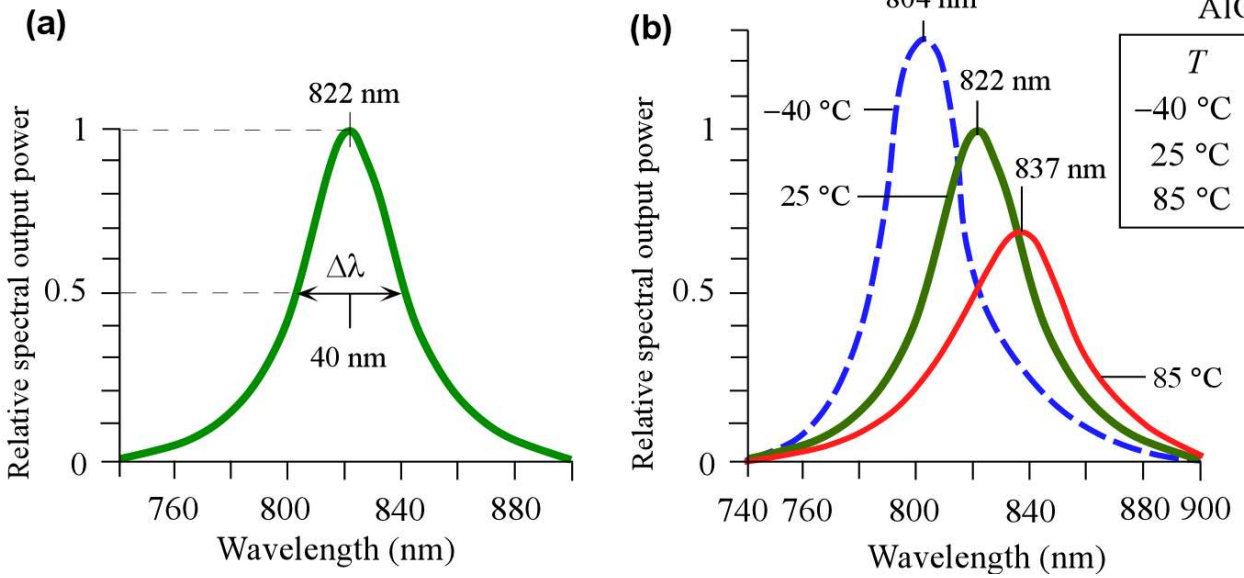
$$\eta_v = (E_v - E_F)/kT$$

$$F_{1/2}(\eta) = \int_0^\infty \frac{\xi^{1/2} d\xi}{1 + e^{\xi - \eta}}$$

$$N_C = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$$

$$N_V = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$$

Emission Spectrum (Exp.)– *symmetrical ?*



AlGaAs IR LED

T	λ_o	$\Delta\lambda$
−40 °C	804 nm	30 nm
25 °C	822 nm	40 nm
85 °C	837 nm	48 nm

$$h\upsilon_o \approx E_g + \tfrac{1}{2}k_BT$$

$$h\Delta\upsilon = mk_BT$$

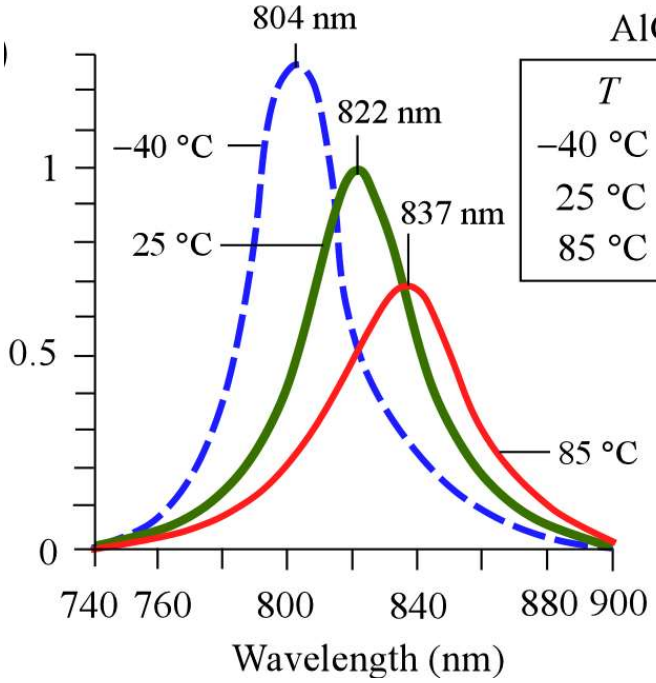
(a) A **typical output spectrum** (relative intensity vs. wavelength)

from an IR (infrared) AlGaAs LED → **Symmetrical ?**

(a) The output spectrum of the LED in (a) **at 3 temperatures**: **25 °C, –40 °C and 85 °C.**

Values normalized to peak emission at 25 °C. The spectral widths are FWHM.

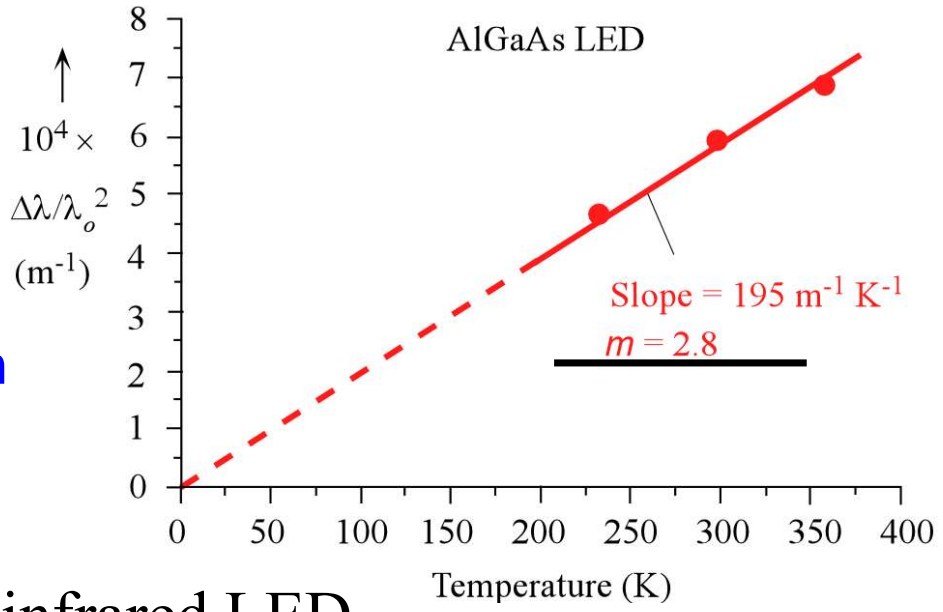
Emission Spectrum



AlGaAs IR LED

T	λ_o	$\Delta\lambda$
−40 °C	804 nm	30 nm
25 °C	822 nm	40 nm
85 °C	837 nm	48 nm

- Output spectrum at 3 temperatures:
  25 °C, –40 °C and 85 °C.
- Values normalized to peak emission at 25 °C.
- Spectral widths: FWHM



- **Energy Gap (T-dependent): Varshni Equation**

$$E_g = E_{go} - AT^2/(B + T),$$

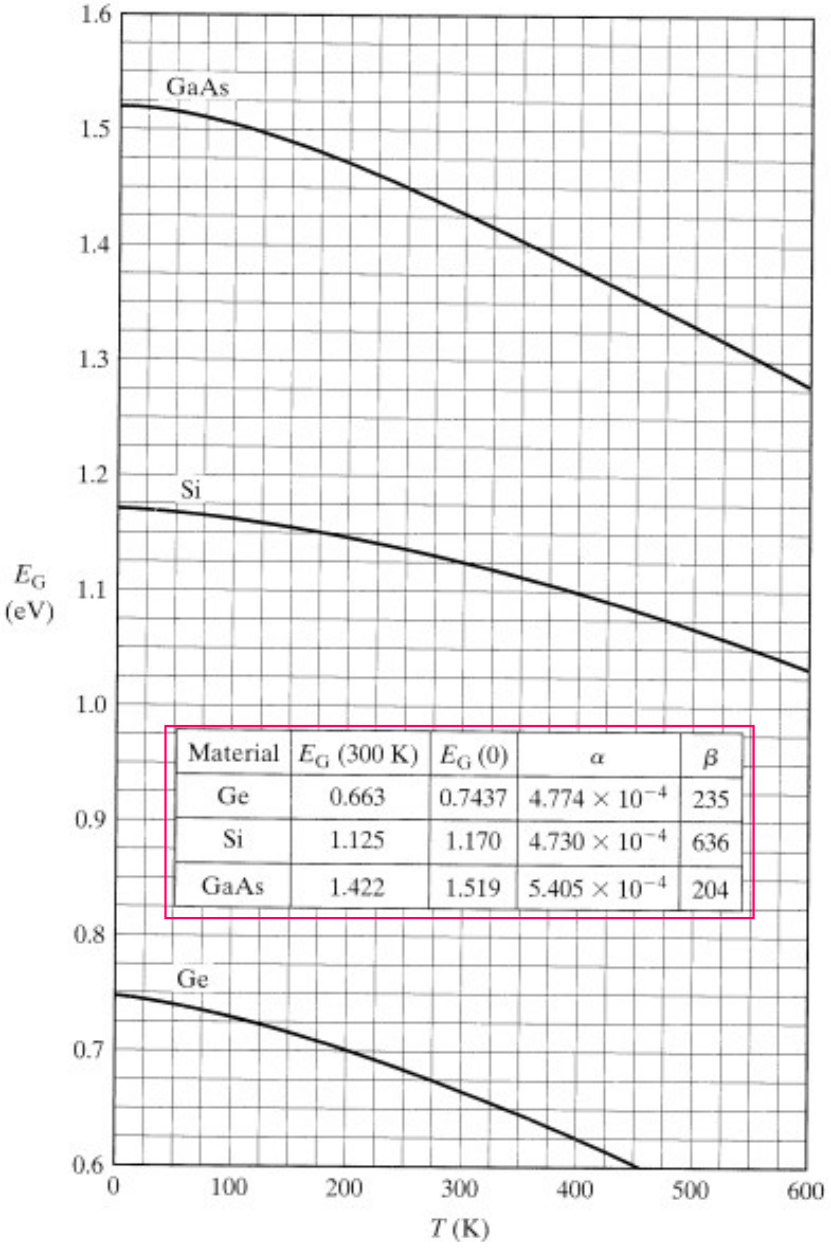
- **The plot of plot $\Delta\lambda/\lambda_o^2$ vs. T for** an AlGaAs infrared LED
  - using the **peak wavelength λ_o**
  & **spectral width $\Delta\lambda$** at three different temperatures

Energy Band Gap: T-dep

$$E_G(T) = E_G(0) - \frac{\alpha T^2}{(T + \beta)}$$

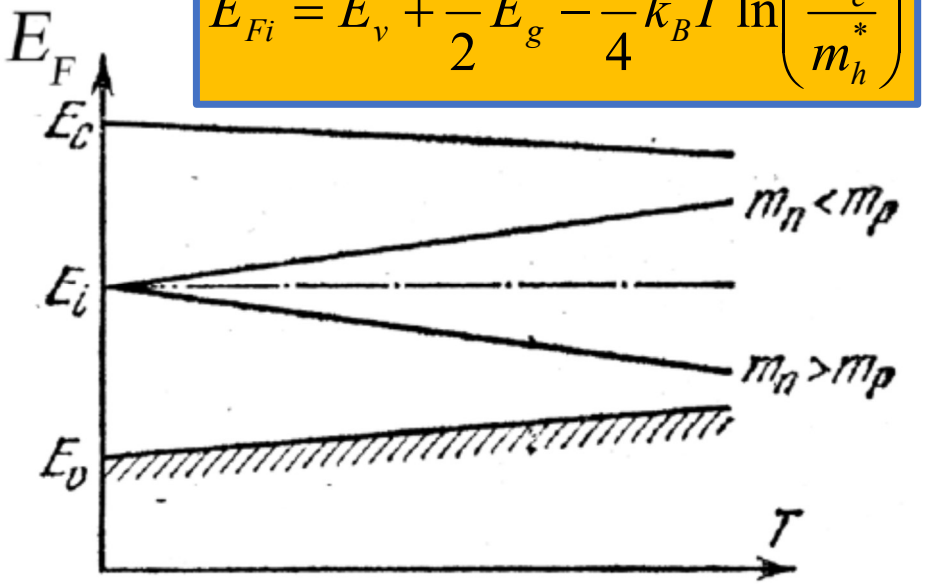
➢ ***Varshni constants*** **for GaAs:**

- E_{go} = 1.519 eV,
- α = 5.41 × 10^{-4} eV K^{-1},
- β = 204 K.

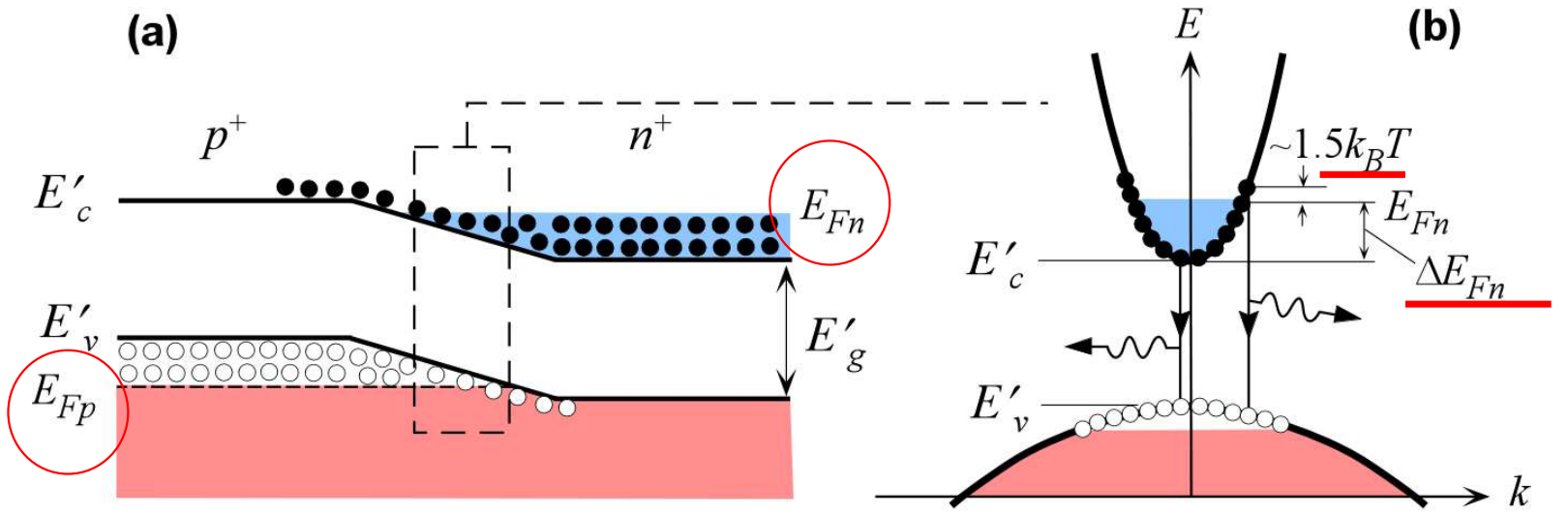


Material	E_G (300 K)	E_G (0)	α	β
Ge	0.663	0.7437	4.774×10^{-4}	235
Si	1.125	1.170	4.730×10^{-4}	636
GaAs	1.422	1.519	5.405×10^{-4}	204

$$E_{Fi} = E_v + \frac{1}{2}E_g - \frac{3}{4}k_B T \ln\left(\frac{m_e^*}{m_h^*}\right)$$



Emission Spectrum-Degenerately Doped pn Junction



(a) **Forward biased degenerately doped *pn* junction.**

- **E'_c is lower than E_c in the bulk**
- **E'_v is higher than E_v in the bulk**
- **The bandgap E'_g is narrower than in the bulk.**

- The quasi-Fermi levels E_{Fn} and E_{Fp} overlap around the junction.

(b) The transitions involved in a degenerately doped *pn* junction.

➢ **Emission spectral width in frequency**

$$\Delta(h\upsilon)$$

$$\approx (E'_g + \Delta E_{Fn} + 1.5\, k_B T) - E'_g$$

$$= \Delta E_{Fn} + 1.5\, k_B T$$

LED spectral linewidth

- **A spread** in the output wavelengths ($\Delta\lambda$)
  - Related to a spread in the emitted photon energies.
- Emitted photon energy: $h\upsilon = hc / \lambda$.
- **When** the spread in the photon energies between the half intensity points: $\Delta(h\upsilon) \approx 3k_BT$.

→ Corresponding linewidth $\Delta\lambda$ between the *half intensity points* in the output spectrum:

LED spectral linewidth

$$\Delta\lambda = \lambda_o^2 \frac{3k_BT}{hc}$$

where λ_o is the peak wavelength.

- **Relationship between the photon frequency υ and λ,**

Photon energy: $E = h\upsilon$ → $\lambda = \frac{c}{\upsilon} = \frac{hc}{h\upsilon}$

→ $\Delta\lambda / \Delta(h\upsilon) \approx |d\lambda / d(h\upsilon)|$ around $\lambda = \lambda_o$

→ **Differentiate** $\frac{d\lambda}{d(h\upsilon)} = -\frac{hc}{(h\upsilon)^2} = -\frac{\lambda^2}{hc}$

→ $$\Delta\lambda = \frac{\lambda_o^2}{hc}\Delta(h\upsilon) = \lambda_o^2 \frac{3k_BT}{hc}$$

Negative sign: Increasing the photon energy decreases the wavelength

$\Delta(h\upsilon) \approx 3k_BT$

EXAMPLE : LED spectral linewidth

(1) Spectral linewidth of an optical communications LED operating at 1310 nm and at 300 K?

where $\Delta(h\upsilon) = 3k_B T$.

$\lambda = 1310$ nm, *and* $T = 300$ K

→ Calculate the linewidth of the 1310 nm LED

$$\Delta\lambda = \frac{\lambda_o^2}{hc}\Delta(h\upsilon) = \lambda_o^2 \frac{3k_B T}{hc}$$

$$\Delta\lambda = \lambda^2 \frac{3k_B T}{hc} = (1310\times10^{-9})^2 \frac{3(1.38\times10^{-23})(300)}{(6.626\times10^{-34})(3\times10^{8})}$$

$= 1.07 \times 10^{-7}$ m or 107 nm

- **Spectral linewidth of an LED output**
  - **Due to the spread in the photon energies** ~ about $3k_B T$.

- **For decreasing $\Delta\lambda$ at a given wavelength**
  → **To reduce the temperature (T).**

✓ **Output spectrum of a laser ~ a much narrower linewidth.**

EXAMPLE: LED spectral width

- Consider the **three experimental points in Figure 3.32 (b) as a function of temperature**. **By** a suitable plot **find *m* and verify**

LED spectral linewidth

$$\Delta\lambda = \lambda_o^2 \frac{3k_B T}{hc}$$

with *m* instead of 3 →

LED linewidth and temperature

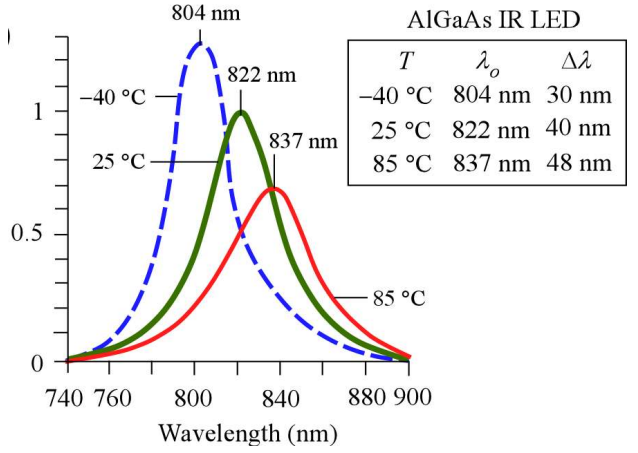
$$\frac{\Delta\lambda}{\lambda_o^2} = \left(\frac{mk_B}{hc}\right) T$$

(1) Plot $\Delta\lambda/\lambda_o^2$ *vs.* *T*.

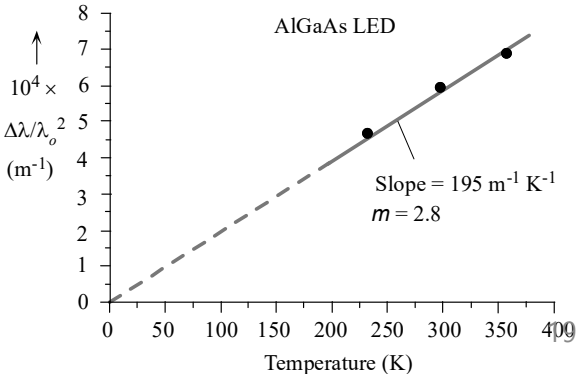
(2) The **slope** of 1.95×10^{-7} nm^{-1} K^{-1} or 195 m^{-1} K^{-1}.

$$\text{slope} = 195 \text{ m K}^{-1} = \frac{m(1.38 \times 10^{-23} \text{ J K}^{-1})}{(6.626 \times 10^{-34} \text{ J s})(3 \times 10^{8} \text{ m s}^{-1})}$$

→ *m* = 2.81



➢ **The plot of plot $\Delta\lambda/\lambda_o^2$ *vs.* *T*** for an AlGaAs infrared LED, using the peak wavelength λ_o and spectral width $\Delta\lambda$ at three different temperatures, using the data shown in the table.



EXAMPLE: Dependence of the emission peak and linewidth on temperature (T)

- **Varshni equation:** $E_g = E_{go} - AT^2/(B + T)$,

  Find the shift in the peak wavelength (λ_o) emitted from a GaAs LED

  when it is cooled from 25 °C to -25 °C.

- ***Varshni constants* for GaAs:** $E_{go} = 1.519$ eV, $A = 5.41 \times 10^{-4}$ eV K^{-1}, $B = 204$ K.

1. At $T = 298$ K, Varshni equation

$E_g = E_{go} - AT^2/(B + T)$

$= 1.519 \text{ eV} - (5.41 \times 10^{-4} \text{ eV K}^{-1})(298 \text{ K})^2/(204 \text{ K} + 298 \text{ K}) =$ **1.423 eV.**

➢ **At 298 K,** $(1/2)\, k_B T = 0.0128$ eV.

✓ **Peak emission:** $h\upsilon_o \approx E_g + (1/2)\, k_B T$.

Using $\upsilon_o = c/\lambda_o$,

$$\lambda_o = \frac{ch}{(E_g + \frac{1}{2} k_B T)} = \frac{(3 \times 10^8 \text{ m s}^{-1})(6.626 \times 10^{-34} \text{ J s}) / 1.602 \times 10^{-19} \text{ eV J}^{-1})}{(1.4223 \text{ eV} + 0.0128 \text{ eV})} = \textbf{864.2 nm}$$

2. At –25 °C or , 248 K, $(1/2)\, k_B T = 0.0107$ eV,

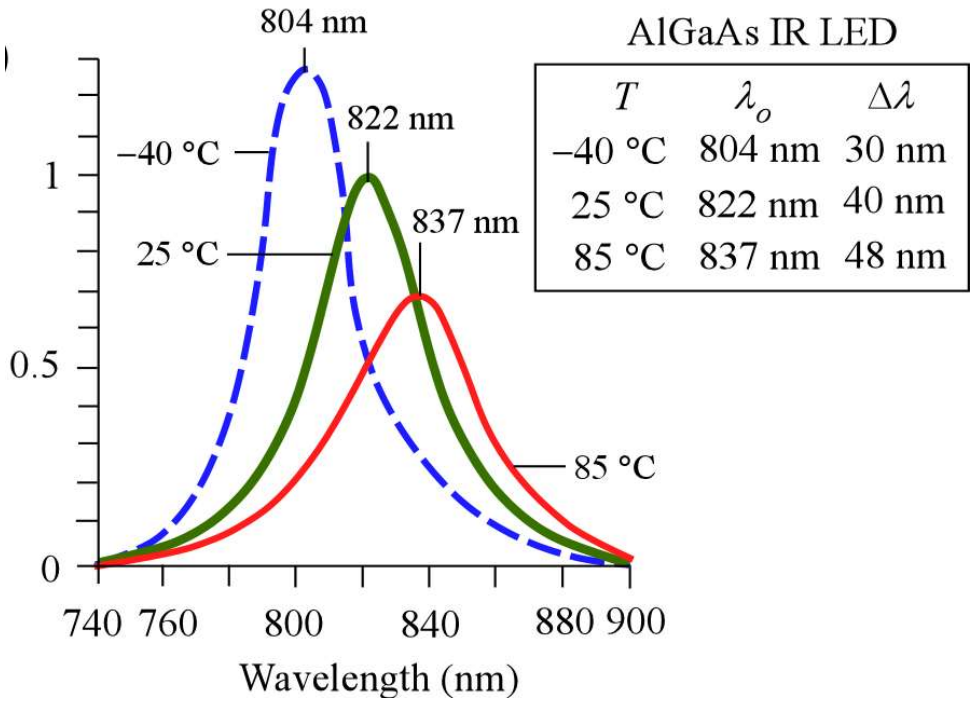
$E_g = 1.519 \text{ eV} - (5.41 \times 10^{-4} \text{ eV K}^{-1})(248 \text{ K})^2/(204 \text{ K} + 248 \text{ K}) =$ **1.445 eV**

EXAMPLE: Dependence of the emission peak and linewidth on temperature (T)

- **New peak emission wavelength λ_0'** **At –25 °C or , 248 K**

$$\lambda_o' = \frac{(3\times10^8\ \mathrm{m\,s^{-1}})(6.626\times10^{-34}\ \mathrm{J\,s})/(1.602\times10^{-19}\ \mathrm{eV\,J^{-1}})}{(1.445\,\mathrm{eV}+0.01069\,\mathrm{eV})} = \mathbf{852.4\ nm}$$

➔ **Change** $\Delta\lambda = \lambda_o - \lambda_0' = 864.2 - 852.4 =$ **11.8 nm** over **50 °C**,
or **0.24 nm / °C**.



AlGaAs IR LED

T	λ_o	$\Delta\lambda$
–40 °C	804 nm	30 nm
25 °C	822 nm	40 nm
85 °C	837 nm	48 nm

➔ **The change in the peak wavelength per unit temperature** in the range –40 °C to 85 °C **is roughly the same (0.24 nm / °C).**